Questions by topic: Soils: fertility, erosion & management - Set 2 (15 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the mean annual soil loss measured on four fields of one farm. Soil forms on this farm at about 1 t ha^{-1} yr^{-1}.

Table 1

Field	Slope	Land use	Mean annual soil loss / t ha^{-1} yr^{-1}
A	2%	Permanent grassland	0.8
B	9%	Maize, ploughed up and down the slope	24.0
C	9%	Maize, contour ploughed	9.0
D	1%	Spring barley, soil left bare over winter	6.5

0 1 . 1 Use Table 1 to identify the fields that are suffering **accelerated** soil erosion, and explain your answer.

[2 marks]

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0 1 . 2 Calculate the percentage reduction in soil loss achieved by changing field B to contour ploughing.

[2 marks]

Answer ..... %

0 1 . 3 Explain how each of the following reduces erosion by water.

[3 marks]

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Contour ploughing .....

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Terracing .....

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Rows of stones .....

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0 1 . 4 Field D is exposed to strong winds. Explain **four** methods, other than planting hedgerows, that would reduce wind erosion there.

[4 marks]

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0 1 . 5 Describe **four** impacts of the eroded soil **away from** the fields where it is lost.

[4 marks]

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END OF QUESTIONS

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